

SANDEEP SHANKAR DHANGE

Full-Stack Developer (React / Node.js)

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PROFILE

Web Engineering Master's student with hands-on full-stack experience building modern web applications using React.js, Node.js, and MongoDB. Delivered production-style projects including an AI-powered system architecture editor built with a five-person team and a large-scale data analytics platform, with a track record of improving performance, designing REST APIs, and automating workflows. Comfortable working across the stack, from responsive UI to backend architecture and database design, in Agile, collaborative environments.

EDUCATION

M.Sc. in Web Engineering

Technische Universität Chemnitz, Germany

Oct 2024 – Present

B.Sc. in Physics, Mathematics & Computer Science (PMCs)

K.L.E. Society's – Raja Lakhamagouda Science Institute, India | CGPA: 8.74

Aug 2020 – Aug 2023

PROJECTS

ArchiFlow — AI-Powered System Architecture Editor (MarxDev)

University Group Project (Team of 5) — MERN Stack, OpenAI API

Nov 2025 – Mar 2026

- Built core full-stack features for an AI-assisted architecture modelling platform: diagram persistence, data management, and property/metadata management for the WAM (WebComposition Architecture Model) editor.
- Designed and implemented REST API endpoints (Node.js/Express, MongoDB) for reliable diagram storage, retrieval, update, and deletion, ensuring accurate reconstruction and secure per-user data association.
- Built the Property Management Panel (React) enabling real-time editing of element attributes, diagram metadata, and save/export workflows, backed by a Jest-based automated test suite.
- Implemented multi-format diagram export (JSON, JSON-LD, Turtle/RDF) to support semantic interoperability and integration with external tools.
- Contributed to frontend canvas development (React Flow) and evaluated diagramming libraries (React Flow, D3.js, Konva.js) to guide the team's technology choice.

German Traffic Accident Analytics Platform

- Built a full-stack MERN platform integrating ~2.1M German traffic accident records (Unfallatlas, 2016–2024) with population data into a harmonised MongoDB schema.
- Designed and documented a REST API (Swagger/OpenAPI) to serve accident trends, regional comparisons, and per-capita rates.
- Developed an interactive React frontend with Recharts and Leaflet for data visualization and map-based exploration of accident hotspots.

WORK EXPERIENCE

Web Developer Intern

Jan 2024 – Jun 2024

- Developed and maintained responsive websites using HTML, CSS, JavaScript, and React, serving live users.
- Built and optimized backend REST APIs with Node.js, improving database query efficiency.
- Partnered with a cross-functional team of developers to design user-friendly dashboards from requirements to delivery.
- Improved website performance and search visibility through code optimization and SEO best practices.
- Applied Git-based version control and Agile methodology in a collaborative team workflow.

SKILLS

Frontend: React.js, JavaScript (ES6+), HTML5, CSS3, Bootstrap, Responsive Design

Backend & Databases: Node.js, Express.js, MongoDB, MySQL, REST API Design

Tools & Practices: Git/GitHub, VS Code, Agile/Scrum, SEO Optimization, Chart.js, Recharts, Leaflet, Swagger/OpenAPI
Languages: English (C1), German (A2)

KEY ACHIEVEMENTS

- Contributed to ArchiFlow, a full-stack AI-powered architecture editor that won 1st place in its final project pitch presentation.
- Led the Decorations Team for college events, managing design, planning, and execution end-to-end (2025).
- Served as Class Representative, acting as the communication bridge between faculty and students (2020).
- Won 1st Prize in a Video Making Competition for creativity and storytelling (2023).
- Secured 3rd Position in a Poster Making Competition for design and visual communication (2023).